Executive Summary: Telecom Customer Churn Analysis

1. Project Overview

The objective of this project is to analyze customer churn in the telecommunications industry and identify customer characteristics, service preferences, and contractual factors associated with customers discontinuing their subscriptions.

The analysis uses exploratory data analysis (EDA) with Python libraries, including Pandas, Matplotlib, and Seaborn. Multiple count plots, a pie chart, a stacked percentage chart, and a tenure histogram were created to explore customer distribution and compare customers who stayed with the company against those who left.

The dataset contains **7,043 customer records** and 21 columns, covering demographic information, service subscriptions, contract types, payment methods, monthly charges, total charges, tenure, and churn status.

The primary objective is to turn these visualizations into actionable insights that can help the business understand customer attrition, identify potentially vulnerable customer segments, and develop more effective retention strategies.

2. Data Preparation and Quality Checks

Several preprocessing steps were completed before the visualization stage:

- **Data type conversion:** The TotalCharges column was converted from text to a numerical data type to support quantitative analysis.
- **Blank-value handling:** Blank entries in TotalCharges were replaced with zero, consistent with the notebook's stated assumption for customers with zero tenure.
- **Missing-value check:** The notebook returned zero missing values after preprocessing.
- **Duplicate checks:** No duplicate rows or duplicate customer IDs were identified.
- **Categorical conversion:** The SeniorCitizen variable was converted from a binary numerical representation into a Yes/No category for easier interpretation.

These steps improved the dataset's usability for exploratory analysis. However, replacing blank charges with zero is an assumption that should ideally be validated against the original data before financial conclusions are drawn.

3. Overall Customer Churn Analysis

The churn count plot and percentage pie chart provide the primary overview of customer retention.

Customer status	Number of customers	Percentage

Stayed with the company	5,174	73.46%
Churned	1,869	26.54%
Total	**7,043**	**100.00%**

Percentages are calculated using the dataset's customer counts.

The results indicate that approximately **26.54% of customers have churned**, while 73.46% have remained with the company. In other words, slightly more than one in four customers in the dataset has discontinued service.

Although most customers remain, the churn proportion is substantial enough to warrant further investigation. The overall figure establishes the baseline against which customer segments can be compared.

From a business perspective, even a modest improvement in retention could be valuable. However, the financial impact of reducing churn would depend on customer lifetime value, acquisition costs, service margins, and the effectiveness of retention initiatives.

4. Customer Demographics and Churn

4.1 Gender-Based Analysis

The gender-wise count plot compares churned and retained customers among male and female customers.

The dataset contains 3,488 female customers and 3,555 male customers. The corresponding churn rates are approximately:

Gender	Customers	Churned customers	Churn rate
Female	3,488	939	26.92%
Male	3,555	930	26.16%

The churn rates differ by less than one percentage point. This suggests that gender alone does not appear to be a strong distinguishing factor in customer churn in this dataset.

Business implication: Retention efforts should not be heavily differentiated by gender based on these results alone. More informative factors, such as contract type, tenure, service configuration, and payment method, deserve greater attention.

4.2 Senior Citizen Analysis

The senior-citizen count plot and the 100% stacked percentage chart examine churn among senior and non-senior customers.

Customer segment	Total customers	Churned customers	Churn rate
Senior citizens	1,142	476	41.68%
Non-senior customers	5,901	1,393	23.61%

Senior citizens account for approximately **16.21% of all customers**, but their churn rate is substantially higher than that of non-senior customers.

The difference is approximately **18.08 percentage points**. Senior customers in this dataset are about 1.77 times as likely to churn as non-senior customers, based on the observed rates.

This is one of the more notable demographic findings in the analysis.

Business implication: The company could investigate whether senior customers experience specific challenges involving pricing, billing, service reliability, product suitability, or customer support. Any retention initiative should be based on evidence about customer needs rather than assumptions about age.

5. Customer Tenure Analysis

The tenure histogram compares the length of customer relationships for customers who churned and those who remained.

The visualization supports an important observation: **customers with shorter tenure appear more vulnerable to churn, while customers with longer tenure are more likely to remain**.

This suggests that the early stages of the customer lifecycle may represent a particularly important period for retention.

Possible explanations include unmet expectations after onboarding, difficulties understanding service features, dissatisfaction with initial billing, or a lack of perceived value. These explanations are hypotheses and are not independently established by the histogram.

Business implication: The company could introduce early-tenure retention initiatives, such as:

- Proactive follow-ups during the first few months.
- Clear explanations of billing and service charges.
- Early identification of service complaints.
- Guidance on using subscribed services.
- Targeted outreach when customers show signs of dissatisfaction.

For a stronger analysis, tenure could also be divided into groups, such as 0–6 months, 7–12 months, 13–24 months, and more than 24 months. Calculating the churn rate for each group would quantify how risk changes as customer relationships mature.

6. Contract Type and Customer Churn

The contract-wise count plot compares churn across month-to-month, one-year, and two-year contracts.

Contract type	Total customers	Churned customers	Churn rate
Month-to-month	3,875	1,655	42.71%
One year	1,473	166	11.27%
Two years	1,695	48	2.83%

Month-to-month customers represent approximately **55.02% of the customer base** and have a churn rate of 42.71%. In contrast, the churn rates for one-year and two-year contracts are approximately 11.27% and 2.83%, respectively.

The month-to-month churn rate is approximately 15 times the two-year contract churn rate. This is a strong association between contract type and customer retention in the observed data.

However, the relationship does not prove that contract duration alone causes churn. Customers choosing longer contracts may already differ in loyalty, price sensitivity, service needs, or other characteristics.

Business implication: Month-to-month customers should be a priority segment for further retention analysis. Potential actions include:

- Offering suitable incentives to move to longer-term contracts.
- Reviewing whether pricing and service bundles provide sufficient value.
- Identifying dissatisfied customers before renewal or cancellation.
- Providing loyalty benefits that reward continued subscription.

Any contract incentives should be evaluated against their cost and their actual impact on customer retention.

7. Service Subscription Analysis

Nine count plots were developed to examine the relationship between churn and the following service variables:

1. Phone service

2. Multiple lines
3. Internet service
4. Online security
5. Online backup
6. Device protection
7. Technical support
8. Streaming TV
9. Streaming movies

These charts provide a broad overview of how service adoption differs between customers who churn and those who stay.

Phone service and multiple lines

The phone-service and multiple-lines plots compare customer retention across basic telephone service configurations. These charts help establish whether customers with different phone-service arrangements display different churn patterns.

The count plots alone do not establish that either service configuration increases churn. Churn rates within each category are needed for a fair comparison.

Internet service

The internet-service chart compares customers using DSL, fiber-optic internet, and those without internet service. It can help identify categories where churn deserves closer examination.

A useful next step would be to calculate the churn rate separately for each internet-service category and investigate whether the results reflect differences in pricing, service quality, customer tenure, or other characteristics.

Online security, online backup, and device protection

These charts compare customers with and without additional protection-related services.

The analysis can help determine whether service adoption is associated with different retention outcomes. If customers without particular services have higher churn rates, the company could investigate whether service bundles, perceived value, or customer needs explain the difference.

Technical support

The technical-support chart is especially useful for exploring whether customers with and without technical support show different churn patterns.

If a higher churn rate is observed among customers without technical support, the company could investigate whether unresolved technical issues contribute to dissatisfaction. However, the chart by itself does not establish this explanation.

Streaming TV and streaming movies

The streaming charts compare customer retention among subscribers and non-subscribers to entertainment services.

These visualizations help distinguish service usage patterns and identify whether particular customer groups merit additional analysis. Streaming subscriptions should not automatically be treated as a retention driver without examining the corresponding churn rates and other customer characteristics.

Overall service-level implication: The nine charts create a useful foundation for understanding service adoption and customer behavior. The next analytical step is to calculate within-category churn rates for all nine variables. This would reveal whether the differences seen in the count plots represent genuine differences in churn risk or simply differences in the number of customers in each category.

8. Payment Method Analysis

The payment-method count plot compares churned and retained customers across electronic checks, mailed checks, bank transfers, and credit cards.

Payment method	Total customers	Churned customers	Churn rate
Electronic check	2,365	1,071	45.29%
Mailed check	1,612	308	19.11%
Bank transfer (automatic)	1,544	258	16.71%
Credit card (automatic)	1,522	232	15.24%

Electronic-check customers have the highest churn rate, at approximately **45.29%**. The corresponding rates for mailed checks, automatic bank transfers, and automatic credit-card payments are considerably lower.

The difference between electronic-check customers and automatic credit-card customers is approximately 30.04 percentage points.

This finding indicates that payment method is a useful variable for further churn investigation. Nevertheless, it does not prove that electronic checks themselves cause customers to leave. Payment method may be associated with other factors, such as contract type, tenure, billing preferences, or customer characteristics.

Business implication: The company could examine whether customers using electronic checks encounter billing friction or differ systematically from customers using automatic payments. It could test whether clearer billing communication or voluntary automatic-payment options improve retention, while avoiding assumptions that a particular payment preference is inherently problematic.

9. Key Business Findings at a Glance

The following table summarizes the strongest numerical patterns identified in the dataset.

Indicator	Observed result	Potential business relevance
Overall churn	26.54%	Establishes the baseline retention rate
Senior-citizen churn	41.68%	Highlights a demographic segment for investigation
Non-senior churn	23.61%	Provides a comparison group
Month-to-month churn	42.71%	Identifies a major contract-related retention concern
One-year contract churn	11.27%	Shows lower observed churn than month-to-month contracts
Two-year contract churn	2.83%	Shows the lowest observed churn among the three contract categories
Electronic-check churn	45.29%	Highlights a payment-method category for further investigation
Female churn	26.92%	Similar to the male churn rate
Male churn	26.16%	Similar to the female churn rate

These are observed associations in the dataset, not proof of causation. The service-level charts and tenure histogram also identify useful areas for additional analysis, but exact category-level churn rates should be calculated before assigning numerical conclusions to them.

10. Recommended Business Actions

Based on the observed patterns, the following priorities are recommended.

Priority 1: Strengthen early-tenure retention. Develop onboarding and customer-engagement initiatives for new subscribers. Monitor cancellations, complaints, and billing issues during the first months of service.

Priority 2: Investigate month-to-month customers. Examine why churn is substantially higher for customers on flexible contracts. Test appropriate loyalty incentives and longer-term plans, and measure whether these interventions improve retention.

Priority 3: Examine electronic-check customers. Compare payment-method groups while controlling for contract type, tenure, and other relevant factors. Investigate billing experiences before introducing payment-related retention initiatives.

Priority 4: Understand senior-customer churn. Investigate customer support, pricing, service suitability, and other potential contributors to the higher observed churn rate among senior citizens.

Priority 5: Improve service-level analysis. Calculate churn rates for phone service, multiple lines, internet service, online security, online backup, device protection, technical support, streaming TV, and streaming movies.

Priority 6: Establish ongoing churn monitoring. Track the overall churn rate and segment-level churn rates over time. Where possible, supplement descriptive analysis with statistical testing or a predictive model to identify customers at elevated risk.

11. Limitations and Further Analysis

This project is an exploratory analysis, and several limitations should be considered:

- The count plots display customer numbers, which can be misleading when category sizes differ. Percentage-based comparisons are necessary.
- Observational associations do not establish causation.
- The notebook does not currently quantify churn rates for every service category.
- The tenure histogram reveals a distributional pattern but does not provide precise churn rates for defined tenure groups.
- No statistical significance tests or predictive churn model are included in the current analysis.
- The handling of blank TotalCharges values should be validated before using the column for financial conclusions.

Future work could include a churn-rate dashboard, tenure-group analysis, a correlation or association study, and a predictive classification model. A multivariable analysis would help establish whether contract type, tenure, payment method, and customer characteristics retain their associations with churn after accounting for one another.

12. Conclusion

This project demonstrates how exploratory data analysis and visualization can be used to investigate customer churn in the telecommunications industry.

The dataset shows an overall churn rate of **26.54%**. The most prominent numerical patterns include higher churn among senior citizens, month-to-month contract customers, and electronic-check customers. Gender-based churn rates are much closer together, indicating that gender alone offers relatively little differentiation in this dataset.

The tenure histogram also suggests that customer retention deserves particular attention during the early stages of the customer relationship. Meanwhile, the nine service-level count plots provide a useful starting point for understanding whether subscribed services are associated with churn.

Taken together, the findings suggest that the most productive next steps are to strengthen early customer engagement, investigate month-to-month contract churn, examine payment-related patterns, and calculate percentage-based churn rates for each service category.

By combining these visualizations with more detailed percentage calculations and further statistical analysis, the project can evolve from descriptive exploration into a practical, evidence-based customer retention strategy.